

Knowledge Distillation and Gradient Estimation for Active Error Compensation in Approximate Neural Networks

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Abstract—Approximate computing is a promising approach for optimizing computational resources of error-resilient applications such as Convolutional Neural Networks (CNNs). However, such approximations introduce an error that needs to be compensated by optimization methods, which typically include a retraining or fine-tuning stage. To efficiently recover from the introduced error, this fine-tuning process needs to be adapted to take CNN approximations into consideration. In this work, we present a novel methodology for fine-tuning approximate CNNs with ultra-low bit-width quantization and large approximation error, which combines knowledge distillation and gradient estimation to recover the lost accuracy due to approximations. With our proposed methodology, we demonstrate energy savings of up to 38% in complex approximate CNNs with weights quantized to 4 bits and 8-bit activations, with less than 3% accuracy loss w.r.t. the full precision model.

Index Terms—Approximate Computing, Neural Networks, Quantization, Approximate multipliers.

I. INTRODUCTION

CNNs play an important role in the field of machine learning, achieving near-human accuracy in tasks such as image classification. However, the implementation of CNN models in edge devices is highly challenging because of large memory requirements, area and power consumption. One paradigm to reduce such requirements is *cross-layer approximate computing*, which comprises approximations at algorithmic and hardware level. In this work, we focus on applying approximate computing to pre-trained CNNs, which facilitates the use of general retraining or fine-tuning schemes that deliver reproducible results, and that can be applied in different application domains. We explore two approximation approaches applied to pre-trained CNNs: Low bit-width quantization at software level, and the use of approximate multipliers at hardware level.

Quantization of CNN parameters to 8 bits has been widely researched. It delivers the same accuracy as the Full Precision (FP) model without additional retraining [1], [2]. However, the use of lower bit-widths can achieve further reduction in memory and power consumption. Therefore, we quantize CNN weights to 4 bits, while maintaining 8-bit activations. We subsequently refer to this configuration as 8A4W quantization.

After applying 8A4W quantization, we incorporate Approximate Multipliers (AMs) in the computation of convolutional

and Fully Connected (FC) layers. The combination of ultra-low bitwidth quantization and AMs makes it more challenging to compensate for the approximation error, as we have less degrees of freedom to “move” to regions where the approximation error is smaller. Furthermore, we are interested in using multipliers with large Mean Relative Error (MRE), which have higher energy savings, but which cause large accuracy degradation. To recover from these approximation errors, a fine-tuning stage is required. This stage consists in applying Stochastic Gradient Descent (SGD) optimization to the approximated CNN for a given number of training epochs. During fine-tuning, the SGD method has to be adapted to take the CNN approximation into account, and thus to perform a better compensation of the approximation error. To this end, in this work we propose a fine-tuning methodology which involves the combination of Knowledge Distillation [3] for approximate CNNs, and the gradient estimation of approximate General Matrix Multiplications (GEMMs). We compare our proposals with two state-of-the-art methodologies: passive retraining [4] and alpha-regularization [5]. In summary, our contributions are:

- Novel two-stage Knowledge Distillation for optimized cross-layer CNN Approximation (ApproxKD).
- Novel gradient estimation of approximate GEMMs for error compensation in approximate CNNs by means of SGD algorithms.
- A methodology combining ApproxKD and gradient estimation for effective error recovery in drastically approximated CNNs.

We perform an extensive analysis of our proposed methodology. For this, we use complex CNNs such as ResNet20, ResNet32 [6] and MobileNetV2 [7] for image classification with CIFAR10 [8]. After model compression through 8A4W quantization, we are able to reach further energy savings of up to 38% with an accuracy loss of 2.37% and 2.36% respectively, when applying approximate multipliers to the evaluated ResNets. We also reach energy savings of 28% with an accuracy loss of 1.76% with the more complex MobileNetV2. To the best of our knowledge, we are the first to successfully approximate such highly complex CNNs after quantizing weights to less than 8 bits.